`GeoModels` Tutorial: simulation, estimation and prediction of positive spatial data using log-gaussian random fields

Moreno Bevilacqua
Christian Caamaño-Carrillo

Introduction

In this tutorial we show how to analyze geo-referenced spatial data with positive support using log-gaussian random fields (RFs) [Oliveira, 2006, Oliveira et al., 1997] with the R package `GeoModels` [Bevilacqua et al., 2026]. In particular, log-Gaussian processes have been widely used for the analysis of positive dependent data due to their well-known mathematical properties. The log-gaussian distribution is a flexible parametric model for positive data allowing right skewness.

We first load the R libraries needed for the analysis and set the name of the model in the `GeoModels` package:

```
rm(list=ls())
require(GeoModels)
require(fields)
model="LogGaussian" # model name in the GeoModels package
```

Simulation of log-Gaussian random fields

The definition of a log-Gaussian RF starts by considering a 'parent' Gaussian RF $Z = \{Z(\boldsymbol{s}), \boldsymbol{s} \in S\}$, where $\boldsymbol{s}$ represents a location in the domain S. In this tutorial, we assume $S = [0,1]^2 \subseteq \mathbb{R}^2$ and that Z is stationary with zero mean, unit variance and correlation function $\rho(\boldsymbol{h}) = \text{cor}\{Z(\boldsymbol{s}+\boldsymbol{h}), Z(\boldsymbol{s})\}$.

Then a RF $V = \{V(\boldsymbol{s}), \boldsymbol{s} \in S\}$ with marginal distribution $LogGaussian(1, \sigma^2)$ can be derived by the transformation

$$V(\boldsymbol{s}) = \exp\{\sigma Z(\boldsymbol{s}) - \sigma^2/2\} \tag{1}$$

where $\sigma > 0$ is a scale parameter. Under this specific parametrization, $\text{E}\{V(\boldsymbol{s})\} = 1$, $\text{var}\{V(\boldsymbol{s})\} = \exp(\sigma^2) - 1$ and the correlation function is given by:

$$\rho_V(\boldsymbol{h}) = \frac{\exp\{\sigma^2 \rho(\boldsymbol{h})\} - 1}{\exp(\sigma^2) - 1}. \tag{2}$$

Then a non-stationary version can be defined through a multiplicative model as:

$$Y(\boldsymbol{s}) = \mu(\boldsymbol{s}) V(\boldsymbol{s}), \qquad \mu(\boldsymbol{s}) > 0. \tag{3}$$

with $\mathrm{E}\{Y(\boldsymbol{s})\} = \mu(\boldsymbol{s})$ and $\mathrm{var}\{Y(\boldsymbol{s})\} = \mu(\boldsymbol{s})^2\{\exp(\sigma^2) - 1\}$. A spatial regression model can be obtained by assuming that $\mu(\boldsymbol{s}) = \exp\{X(\boldsymbol{s})^T\boldsymbol{\beta}\}$, where $X(\boldsymbol{s})$ is a k-dimensional vector of covariates and $\boldsymbol{\beta} = (\beta_0, \ldots, \beta_{k-1})^T$ is a k-dimensional vector of unknown parameters.

Thus, in order to obtain a realization from a log-Gaussian RF we need to specify regression mean parameters, a scale parameter and a parametric correlation model for $\rho(\boldsymbol{h})$. We first set the spatial coordinates:

```
N=1500 # number of location sites
set.seed(89)
x = runif(N,0,1)
y = runif(N,0,1)
coords=cbind(x,y)
```

Then we fix $k = 2$ and we build the matrix covariates and fix the regression mean parameters:

```
X=cbind(rep(1,N),runif(N)) # matrix covariates
mean=-0.2; mean1=0.25 # regression parameters
nugget=0 # nugget parameter
sill=1
```

where `mean` and `mean1` are respectively β_0 and β_1. The `sill` parameter of the log-Gaussian RF and the `nugget` parameter are respectively σ^2 and τ.

The names of the marginal parameters associated with the log-Gaussian model can be obtained with the function `NuisParam` (note that the option `num_betas` is the number of regression parameters involved):

```
NuisParam(model,num_betas=2)
# [1] "mean" "mean1" "nugget" "sill"
```

For the correlation function we assume a special case of the isotropic Generalized Wendland class [Bevilacqua et al., 2019], i.e. the Askey model:

$$\rho(\boldsymbol{h}; \alpha, \delta) = \begin{cases} (1 - \|\boldsymbol{h}\|/\alpha)^{\delta}, & \|\boldsymbol{h}\| < \alpha, \\ 0, & \text{otherwise.} \end{cases}$$

Using asymptotic arguments Bevilacqua et al. [2019] show that this correlation model has the same local regularity features as the exponential correlation model. Additionally it is compactly supported, an interesting feature from the computational point of view. We set

the Askey model and the associated parameters. Note that the function `CorrParam` returns the names of parameters associated with a given correlation model:

```
corrmodel="Wend0" # correlation model and parameters
CorrParam("Wend0")
# [1] "power2" "scale"
scale=0.25
power2=4
```

Here the `scale` parameter corresponds to α, the compact support of the correlation model. We are now ready to simulate a log-Gaussian random field using the function `GeoSim`:

```
param=list(mean=mean,mean1=mean1,sill=sill,nugget=nugget,
          scale=scale,power2=power2)
data=GeoSim(coordx=coords,corrmodel=corrmodel,model=model,
           param=param,X=X)$data
```

The simulation is performed using Cholesky decomposition.

Estimation of log-Gaussian random fields

The density of the bivariate random vector $(V(\boldsymbol{s}_i), V(\boldsymbol{s}_j))^T$ is given by

$$f_V(v_i, v_j) = \frac{1}{2\pi\sigma^2 v_i v_j \sqrt{1-\rho^2(\boldsymbol{h})}} \times \exp\left\{-\frac{1}{2\{1-\rho^2(\boldsymbol{h})\}}\left[\left(\frac{\log v_i}{\sigma}+\frac{\sigma}{2}\right)^2+\left(\frac{\log v_j}{\sigma}+\frac{\sigma}{2}\right)^2 -2\rho(\boldsymbol{h})\left(\frac{\log v_i}{\sigma}+\frac{\sigma}{2}\right)\left(\frac{\log v_j}{\sigma}+\frac{\sigma}{2}\right)\right]\right\}. \tag{4}$$

and the bivariate densities of Y can be derived from (4) as

$$f_Y(y_i, y_j) = (\mu_i\mu_j)^{-1} f_V(y_i/\mu_i, y_j/\mu_j). \tag{5}$$

Given a realization $y(\boldsymbol{s}_1), \ldots, y(\boldsymbol{s}_N)$ of Y, the conditional pairwise log-likelihood function is defined as:

$$pl(\boldsymbol{\theta}) = \sum_{i=1}^{N}\sum_{\substack{j=1\\ j\neq i}}^{N} \log\left\{\frac{f_Y(y(\boldsymbol{s}_i), y(\boldsymbol{s}_j))}{f_Y(y(\boldsymbol{s}_j))}\right\} w_{ij},$$

where w_{ij} are non-negative weights, not depending on $\boldsymbol{\theta}$. An efficient way to specify the weights from a computational viewpoint is based on neighborhoods:

$$w_{ij}(k) = \begin{cases} 1, & \boldsymbol{s}_i \in N_k(\boldsymbol{s}_j), \\ 0, & \text{otherwise.} \end{cases} \tag{6}$$

Here $N_k(\boldsymbol{s}_l)$ is the set of the k nearest neighbors of $\boldsymbol{s}_l$, and in this case $\boldsymbol{\theta} = (\beta_0, \beta_1, \sigma^2, \alpha, \delta)^T$. The conditional pairwise likelihood estimator $\widehat{\boldsymbol{\theta}}_{pl}$ is obtained by maximizing $pl(\boldsymbol{\theta})$ with respect to the free parameters. In the `GeoModels` package we can choose the fixed parameters and the parameters that must be estimated. Conditional pairwise likelihood estimation is performed with `GeoFit2`:

```
start=list(mean=mean,mean1=mean1,scale=scale,sill=sill)
fixed=list(nugget=nugget,power2=power2)
# Maximum composite-likelihood fitting of the loggaussian RF:
fit=GeoFit2(data=data,coordx=coords,corrmodel=corrmodel,
     model=model,X=X,likelihood="Conditional",type="Pairwise",
     start=start,fixed=fixed,neighb=1)
```

The option `neighb=1` sets the k value (i.e. the neighborhood order) in (6). The object `fit` includes information about the conditional pairwise likelihood estimation:

```
fit
# Original tutorial output (the values have not been recalculated):
# Maximum Composite-Likelihood Fitting of Log Gaussian Random Fields
# Setting: Conditional Composite-Likelihood
# Model: LogGaussian
# Type of the likelihood objects: Pairwise
# Covariance model: Wend0
# Optimizer: Nelder-Mead
# Number of spatial coordinates: 1500
# Number of estimated parameters: 4
# Type of convergence: Successful
# Maximum log-Composite-Likelihood value: -144.02
# Estimated parameters:
# mean mean1 scale sill
# -0.3310 0.2640 0.2794 1.0579
```

Checking model assumptions

Given the estimation of the mean $\widehat{\mu}(\boldsymbol{s}) = \exp\{X_1(\boldsymbol{s})\widehat{\beta}_0 + X_2(\boldsymbol{s})\widehat{\beta}_1\}$, the estimated residuals

$$\widehat{v}(\boldsymbol{s}_i) = \frac{y(\boldsymbol{s}_i)}{\widehat{\mu}(\boldsymbol{s}_i)}, \quad i = 1, \ldots, N, \tag{7}$$

can be viewed as a realization of V, a stationary RF with log-Gaussian marginal distribution, unit mean and correlation function given by (2). The estimated residuals can be computed using `GeoResiduals`:

```
res=GeoResiduals(fit) # computing residuals
```

Then the agreement of the marginal distribution assumption on the residuals with the theoretical model can be graphically checked with the `GeoQQ` function (Figure 1, left part):

```
GeoQQ(res,pch=20)
```

The covariance model assumption can be checked comparing the empirical and the estimated semivariogram using the `GeoVariogram` and `GeoCovariogram` functions (Figure 1, right part):

```
vario=GeoVariogram(data=res$data,coordx=coords,maxdist=0.3)
GeoCovariogram(res,show.vario=TRUE,vario=vario,pch=20,
               ylim=c(0,2.5))
```

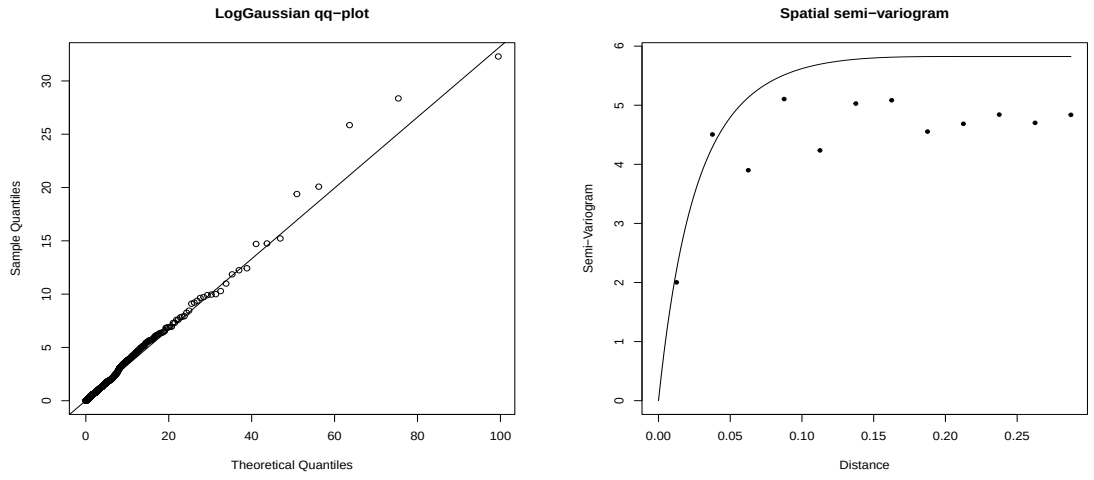


Figure 1: Left: QQ-plot for the log-Gaussian model residuals. Right: empirical vs estimated semi-variogram function for the residuals.

Prediction of log-Gaussian random fields

The optimal linear prediction of a log-Gaussian RF at a location $\boldsymbol{s}_0$ is given by (see Oliveira [2006] and Bevilacqua et al. [2020] for related prediction methods):

$$\widehat{Y}(\boldsymbol{s}_0) = \widehat{\mu}(\boldsymbol{s}_0)\left(1 + \sum_{i=1}^{N} \widehat{\lambda}_i[\widehat{V}(\boldsymbol{s}_i) - 1]\right), \tag{8}$$

where the vector of weights $\boldsymbol{\lambda} = (\lambda_1, \ldots, \lambda_N)^T$ is given by $\boldsymbol{\lambda} = \boldsymbol{R}^{-1}\boldsymbol{c}$. Here the cross-correlation vector and the correlation matrix, respectively, are

$$\boldsymbol{c} = \big(\text{cor}\{V(\boldsymbol{s}_0), V(\boldsymbol{s}_1)\}, \ldots, \text{cor}\{V(\boldsymbol{s}_0), V(\boldsymbol{s}_N)\}\big)^T, \qquad \boldsymbol{R} = [\text{cor}\{V(\boldsymbol{s}_i), V(\boldsymbol{s}_j)\}]_{i,j=1}^{N}.$$

Both are computed using the correlation function (2). We first set the spatial locations to predict and the associated covariates. In this example, we choose a regular fine grid in order to construct a prediction map:

```
xx=seq(0,1,0.013)
loc_to_pred=as.matrix(expand.grid(xx,xx))
Nloc=nrow(loc_to_pred)
Xloc=cbind(rep(1,Nloc),runif(Nloc))
```

Then the optimal linear prediction (8), using the estimated parameters, can be performed using the `GeoKrig` function:

```
pr=GeoKrig(fit,loc=loc_to_pred,Xloc=Xloc,
                    mse=TRUE,sparse=TRUE)
```

We can compare the map of simulated data with the kriging prediction (and associated mean square error) with the following code (see Figure 2):

```
par(mfrow=c(1,3))
quilt.plot(x,y,data,main="Data")
map=matrix(pr$pred,ncol=length(xx))
image.plot(xx,xx,map,xlab="",ylab="",main="Kriging")
map_mse=matrix(pr$mse,ncol=length(xx))
image.plot(xx,xx,map_mse,xlab="",ylab="",main="MSE")
```

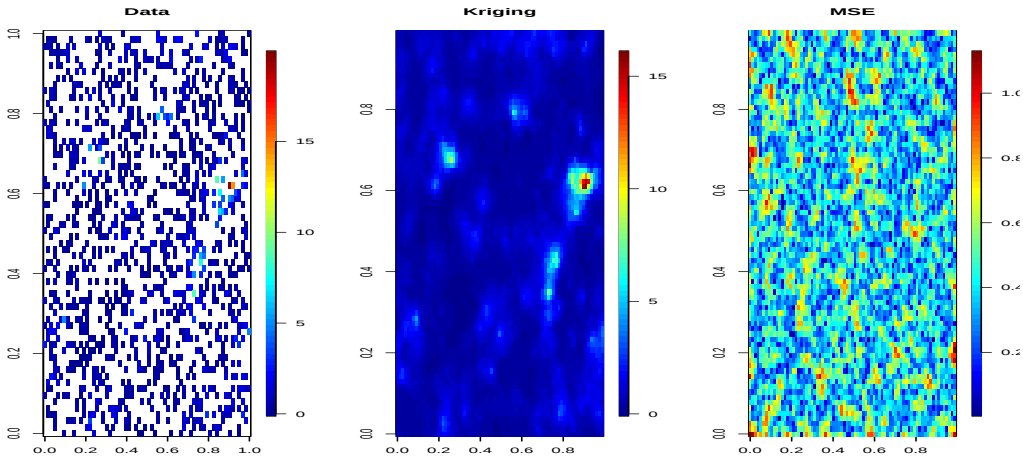


Figure 2: From left to right: colored map of observed data, kriging prediction and associated mean squared error.

References

Moreno Bevilacqua, Tarik Faouzi, Reinhard Furrer, and Emilio Porcu. Estimation and prediction using generalized Wendland covariance functions under fixed domain asymptotics. *The Annals of Statistics*, 47(2):828–856, 2019. doi: 10.1214/17-AOS1652.

Moreno Bevilacqua, Christian Caamaño-Carrillo, and Carlo Gaetan. On modeling positive continuous data with spatiotemporal dependence. *Environmetrics*, 31(7):e2632, 2020. doi: 10.1002/env.2632.

Moreno Bevilacqua, Christian Caamaño-Carrillo, Francisco Cuevas-Pacheco, and Víctor Morales-Oñate. *GeoModels: Procedures for Gaussian and Non Gaussian Geostatistical (Large) Data Analysis*, 2026. URL `https://cran.r-project.org/package=GeoModels`. R package version 2.3.0.

Victor De Oliveira. On optimal point and block prediction in log-Gaussian random fields. *Scandinavian Journal of Statistics*, 33(3):523–540, 2006. doi: 10.1111/j.1467-9469.2006.00494.x.

Victor De Oliveira, Benjamin Kedem, and David A. Short. Bayesian prediction of transformed Gaussian random fields. *Journal of the American Statistical Association*, 92(440): 1422–1433, 1997. doi: 10.1080/01621459.1997.10473663.